

CWF: A Collaborative Writing Framework for Personalized and Reliable Popular Science Writing

Ruibiao Fu^{1,†}, Di Tang^{1,†}, Yunlong Yang^{1,†}, Ran Wang^{1,2,*}, Sicheng Lu¹

Peixuan Wu¹, Xiaoyu Fan¹, Jiacheng Ma¹, Haozhe Luo¹, Yang Xiao³

¹ School of Journalism and Information Communication, Huazhong University of Science and Technology, Wuhan 430074, China

² Philosophy and Social Sciences Laboratory of Big Data and National Communication Strategy, Ministry of Education, Wuhan 430074, China

³ The National Key Laboratory of Multispectral Information Intelligent Processing Technology, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China

{furuibiao3, tangdi030223, yunlongyang64, xiaoyuf30}@gmail.com

{rex_wang, loo_seychan, w_px, jiacheng_ma, luo_haozhe, Yang_Xiao}@hust.edu.cn

Abstract

We introduce **Personalized and Reliable Popular Science Writing**, a novel task that requires adapting scientific explanations to audiences with different cognitive levels while preserving factual accuracy. However, improving personalization often introduces simplifications that increase the risk of hallucination and factual distortion. To address these challenges, we first construct a dataset of 39,134 entries and a reader-centric **Personalized Science Communication Benchmark (PSCB)** that jointly evaluates audience adaptation and factual accuracy. To reduce data and computational requirements while improving generalization across domains and audiences, we introduce DA-MoE, which explicitly decouples audience adaptation from domain knowledge through separate modeling. To enable robust verification and revision in evidence-scarce scenarios, a multi-agent fact-checking mechanism that augments limited evidence with role-specific agent debate and propagates confidence over a graph is proposed. Experiments on PSCB show that our approach achieves state-of-the-art performance. Our code is open-sourced at <https://github.com/DPIInnovationWorks/CWF>.

1 Introduction

Popular science writing aims to communicate scientific knowledge to audiences in an accurate and accessible way. Large language models (LLMs) have introduced new possibilities for this task by

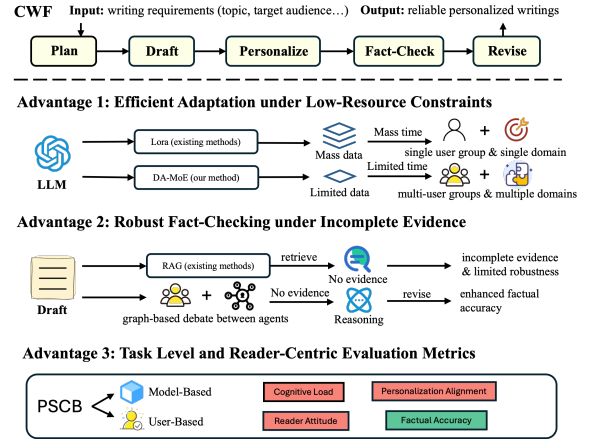


Figure 1: Comparison between Specialized Approaches (Fine-tuning and RAG-based methods) and Collaborative Writing Framework (CWF).

enabling flexible text generation across domains and writing styles. However, effective popular science writing requires more than fluent generation. The generated article should be adapted to the target audience in both writing style and cognitive adaptation, while remaining factual accuracy. Existing methods mainly rely on prompt engineering (Kim et al., 2024; Tang et al., 2024), retrieval-based verification (Min et al., 2023a), or supervised fine-tuning (SFT) (Goldsack et al., 2022). Despite their effectiveness, these approaches usually treat writing style, audience adaptation, and factual accuracy as separate objectives, making it difficult to generate articles that are both personalized and reliable.

This limitation becomes more pronounced in SFT. High-quality popular science writing often

[†]The three authors contribute equally to this work.

^{*}Corresponding author.

requires jointly aligned data that couples domain knowledge, audience adaptation, and writing style (Guo et al., 2022). Such data is costly to collect because it must contain both accurate knowledge and audience-adaptive expressions. Moreover, as a generative task, popular science writing may produce factual omissions and hallucinated explanations. Therefore, the key challenge is not only to reduce the dependence on large-scale paired data for domain and audience adaptation, but also to ensure factual accuracy throughout the generation process.

Existing benchmarks do not adequately address this challenge, they typically emphasize stylistic preference (August et al., 2024; Tang et al., 2024) or factual accuracy in popular science writing (Min et al., 2023b; Damm et al., 2024; Anonymous, 2025), but rarely consider both of them within a unified benchmark. Audience adaptation often changes explanation granularity, analogy use, implicit reasoning, and conceptual simplification, which may increase the risk of unsupported information or factual distortion (Devaraj et al., 2022; Taylor and Dewsbury, 2018). Accordingly, PSCB benchmark is introduced for personalized and reliable popular science writing, as shown in Figure 1. PSCB jointly evaluates cognitive adaptation and factual accuracy. Beyond reader attitude and personalization alignment, PSCB first introduces Cognitive Load (Sweller, 1988; Salemi et al., 2023; Ouwehand et al., 2021; Zu et al., 2021) as a metric and decomposes metrics into fine-grained dimensions for assessing cognitive adaptation beyond surface style.

To address the challenges of costly jointly aligned data and the difficulty of balancing personalization with factual accuracy, we design the Collaborative Writing Framework (CWF). We first introduce a decoupled MoE architecture for popular science writing. This framework relaxes the requirement for paired data that contains the target audience and scientific knowledge. This design allows us to construct a dataset of 39,134 entries that does not rely on joint alignment between domains and audiences, enabling training across different scientific domains and reader groups. However, simple MoE faces difficulty when transferring to new domains, since generating articles in different domains may require retraining the whole model. We propose DA-MoE, whose key idea is to assign adaptation tasks to individual pluggable experts and to separate audience-specific adaptation from

domain-specific knowledge, so that cognitive adaptation and knowledge can be learned from different sources. This allows to adapt to new domains with only lightweight training.

Popular science writing requires more than cognitive adaptation. Even well-personalized generations may contain omissions, unsupported claims, or hallucination, making factual verification essential. CWF therefore incorporates a fact-checking module to preserve factual accuracy during generation. However, RAG-based methods struggle when direct evidence is sparse or conflicting because they rely primarily on retrieved evidence rather than deriving additional support by reasoning over indirect viewpoints or evidences (Asai et al., 2023). Moreover, most post-hoc fact-checking methods rely on direct model judgments over completed outputs, not only producing unquantifiable results but also cannot guide the writing process (Jiang et al., 2023; Min et al., 2023a; Zheng et al., 2023; Wei et al., 2024). Furtherly, we propose a verification mechanism that combines multi-agent debate with graph-based inference. Specifically, multi-agent debate simulates an interactive review process, and helps validate claims by introducing complementary perspectives and broadening retrieval angles when direct evidence is incomplete. Intervention applies verification during the generation process rather than only after generation, allowing multi-agent factual feedback to guide the writing process itself. Finally, graph-based inference connects claims, retrieved evidence, and agent reasoning, enabling additional evidence to be inferred from indirect viewpoint while making factual judgments traceable rather than black-box decisions.

In summary, **the main contributions** of this work are as follows:

- Personalized and reliable popular science writing task is proposed for the first time. We further introduce PSCB, a human-aligned benchmark that jointly evaluates audience adaptation and factual accuracy.
- We propose a DA-MoE framework for popular science writing that decouples domain knowledge from audience adaptation, reduces reliance on paired training data, and enables low-cost generalization to new domains.
- We propose a multi-agent fact-checking mechanism with graph-based reasoning. It simulates the debate process to support multi-

perspective verification, intervention during generation, and quantifiable confidence scores for robust claim validation.

2 Related Works

Popular Science Writing. Recent work in computational linguistics studied how to transform scientific content into language that general audiences can understand. Goldsack et al. (Goldsack et al., 2022) and Cheng et al. (Cheng et al., 2025) formulate this problem as lay summarization or academic-to-public paraphrasing by constructing paired datasets and benchmarks, including PLOS, eLife, and VTechAGP. Fang et al. (Fang et al., 2024) and Jiang et al. (Jiang et al., 2025) further study reliability in popular science generation by analyzing factual errors in LLM-generated biomedical summaries or using multi-agent interaction to improve science journalism generation. However, existing studies usually address cognitive adaptation and factual accuracy separately. They do not provide a unified framework that jointly considers personalization for different readers and factual accuracy in popular science writing. To bridge this gap, we formulate personalized and reliable popular science writing and introduce PSCB to jointly evaluate reader adaptation and factual accuracy.

Personalized and Controllable Writing. Personalized generation primarily utilizes Few-Shot Prompting or SFT (Houlsby et al., 2019; Brown et al., 2020; Li and Liang, 2021). However, prompt-based methods often struggle with long-form text, while fine-tuning imposes strict data requirements to model target styles effectively. This limitation is particularly pronounced when combining heterogeneous styles, which requires constructing costly datasets that jointly capture multiple stylistic patterns (Ju and Wang, 2024). To reduce this dependence, we propose DA-MoE, which decouples audience adaptation from domain knowledge and enables flexible composition across reader groups and scientific domains.

Fact-checking in LLMs. Traditionally, fact-checking has heavily depended on human evaluators, a process that is not only time-consuming and expensive but also prone to subjective biases (Honovich et al., 2022; Devaraj et al., 2022; Min et al., 2023a). Subsequently, approaches combining LLMs with RAG emerged to integrate external databases (Asai et al., 2023). However, they often struggle when direct evidence is sparse or conflict-

ing, due to a lack of deep reasoning capabilities. Recent methods have proposed decomposing complex claims into atomic units (Min et al., 2023a; Wei et al., 2024) or simulating multi-agent debates (Sun et al., 2025; Ma et al., 2026). By supplementing the original claims with additional context, these approaches enable judgments from multiple perspectives. However, they remain post-hoc and static, unable to instruct in the generation process proactively. Also, while simulated multi-agent debates introduce interactive reasoning, they rely on model judgments without providing rigorous metrics. To address these issues, we propose a verification framework. By combining multi-agent debate with graph, our method not only synthesizes diverse perspectives to handle incomplete evidence but also provides quantifiable confidence scores to individual reasoning nodes, ensuring a transparent verification process.

Evaluation Benchmark. Existing benchmarks either emphasize readability and audience accessibility in science communication (August et al., 2024; Tang et al., 2024) or focus on factual accuracy and factuality evaluation in generated texts (Min et al., 2023b,a; Damm et al., 2024; Anonymous, 2025; Wei et al., 2024). However, these two lines of work are usually studied separately, making them insufficient for evaluating popular science generation across different audiences. Our Personalized Science Communication Benchmark (PSCB) evaluates whether a generated science article is both reliable and adapted to its intended readers. In real-world science communication, an effective article should not only convey scientifically grounded knowledge, but also adjust its conceptual difficulty, explanation strategy, and communicative style to different reader groups. PSCB therefore contains two complementary evaluation branches: Factual Reliability and Audience Adaptation.

3 Collaborative Writing Framework

As illustrated in Figure 2, the proposed CWF aims to generate coherent science writing that remain both personalized and factually reliable. CWF organizes the writing into three coordinated stages—drafting, personalization, and verification—to address the core challenge of balancing personalization and factual accuracy.

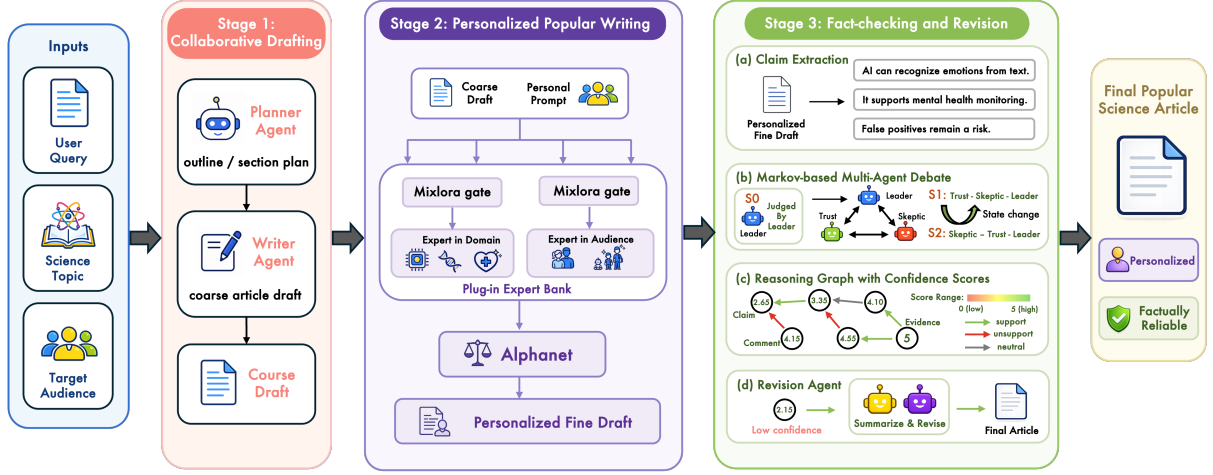


Figure 2: Overview of CWF, which integrates collaborative writing, personalized style modeling, and fact-checking with multi-agent discussion to enhance the quality and relevance of AI-generated science writings.

3.1 Collaborative Writing

Existing LLM-based popular science writing methods suffer from structural inconsistency, weak coherence, and limited control over narrative organization (Shao et al., 2024; Jiang et al., 2024). To address this, we adopt a collaborative generation framework in which one agent plans the article structure and section summaries, while another expands them into fluent passages with examples and transitions. This process forms the initial draft and improves narrative coherence.

3.2 DA-MoE

To address the difficulty of data collection in SFT, we build a modular MoE-based generation framework on top of the plug-and-play MixLoRA architecture (Li et al., 2024). The framework consists of two specialized MoE networks constructed from multiple Qwen2.5-3B backbone models: an audience-adaptation MoE for learning audience-specific writing styles, and a knowledge-domain MoE for learning topic-specific knowledge.

Formally, given an input x , we construct two MoE modules: an audience-adaptation MoE and a knowledge-domain MoE. Let $r \in \{s, k\}$ denote the expert type, where s represents audience adaptation and k represents knowledge domain. The corresponding expert sets are defined as:

$$\mathcal{E}^r = \{E_1^r, E_2^r, \dots, E_{T_r}^r\}, \quad (1)$$

where $T_s = M$ and $T_k = N$. Each expert is initialized from Qwen2.5-3B. Audience experts are fine-tuned on data associated with specific audience

adaptation, while knowledge experts are fine-tuned on data from specific scientific domains.

For each MoE module, a lightweight MixLoRA gating network assigns expert weights based on the input. The routing distribution is computed as:

$$g^r(x) = \text{softmax}(\text{MLP}_r(x)), \quad (2)$$

where MLP_r denotes the gating network for expert type r . The output of each MoE module is then:

$$h^r(x) = \sum_{i=1}^{T_r} g_i^r(x) E_i^r(x), \quad (3)$$

where $g_i^r(x)$ is the routing weight assigned to the i -th expert in $\mathcal{E}^r$.

The training process has three stages. First, each Qwen2.5-3B expert is independently fine-tuned on its assigned style or knowledge data, so that the style experts specialize in audience adaptation and the knowledge experts specialize in domain-specific scientific content. Second, after the expert parameters are fixed, we train the MixLoRA gating networks using a small amount of supervised data, allowing the model to learn how to route each input to appropriate style and knowledge experts. Third, we introduce an AlphaNet module to dynamically balance the contributions of the two MoE networks during generation. Specifically, AlphaNet is also parameterized as an MLP and directly predicts an input-dependent coefficient:

$$\alpha = \text{MLP}_{\text{alphanet}}(x). \quad (4)$$

The final hidden representation is then obtained by contrasting the style-oriented and knowledge-

Reader Group	Mean	Std.	Median	Min	Max
Children	0.686	0.101	0.697	0.383	0.846
Teenagers	0.262	0.080	0.256	0.095	0.480
Adults	0.183	0.059	0.176	0.079	0.351

Table 1: Distribution of α across different target reader groups ($n = 100$ per group).

Method	Training Time	Data Amount	Data Type	Average PSCB score
LoRA	2.1 hrs	5000	Dual-style	2.985
MoE	8.3 hrs	20000	Single-style	3.059
DA-MoE	22 min	1000	Single-style	3.451

Table 2: Comparison of training time, data requirements, data type, and average PSCB score in different methods.

oriented MoE outputs:

$$h(x) = \alpha * h^s(x) + (1 - \alpha) * h^k(x). \quad (5)$$

To examine whether AlphaNet learns different balances for different target audiences, we analyze the distribution of α across reader groups.

As shown in Table 1, the mean α decreases from children (0.686) to teenagers (0.262) and adults (0.183). Since a larger α indicates greater reliance on the style network, this result suggests that the model places more emphasis on style and readability for younger readers, while relying more on knowledge content for adult readers. This pattern is also consistent with previous studies showing that narrative and style-oriented presentation can improve comprehension and engagement for non-expert audiences (Dahlstrom, 2014; Downs, 2014).

Table 2 shows that our method achieves the highest average PSCB score while requiring substantially less training time and fewer training examples. This efficiency mainly comes from the modular design of our framework. When extending the model to a new domain, we do not need to collect large-scale dual-style data or retrain the full model. Instead, only a small amount of single-style data is required to train the gating network, while the existing style and knowledge experts can be reused.

3.3 Fact-checking and Revision

Our framework performs fact-checking through a multi-agent collaboration that simulates interactive discussions around scientific claims. Rather than relying on a single model to assess factuality, we decompose the verification process into a sequence of role-specific steps.

Claim Extraction. Given a draft article, the system parses each paragraph into a set of atomic claims, ensuring that each statement can be independently verified. For each decomposed claim, we estimate its coverage to the original paragraph and the sums of all coverage scores are 1, which are used as weights in confidence aggregation.

Markov Chain-Based Debate. We instantiate three complementary roles: Leader acts as an expert who gives the scientific judgment; Trust agent acts as a reader-oriented supporter who searches for reasonable evidence and complementary explanation; Skeptic agent acts as a critics who challenges unsupported claims and weak evidence. Science communication is not merely a one-way transmission of expert knowledge, but a mediated process involving expert judgment, audience-oriented interpretation, and critical gatekeeping (Wynne, 1992; Burns et al., 2003; Fahy and Nisbet, 2011). Accordingly, the Leader, Trust, and Skeptic agents respectively operationalize these three functions for factual verification, with detailed role designs provided in Appendix B.

Given an atomic claim, the process starts from S_0 , where the Leader provides an initial judgment. The subsequent debate order is controlled by the factuality signal $f(L_t)$ parsed from the latest Leader output, where $f(L_t) = 1$ for supportive stances (1, 0.2) and $f(L_t) = 0$ for neutral or opposing stances (0, -0.2, -1):

$$S_{t+1} = \begin{cases} S_2, & f(L_t) = 1, \\ S_1, & f(L_t) = 0. \end{cases} \quad (6)$$

When Leader initially supports a claim, Skeptic is invoked first to prevent premature acceptance; when Leader is uncertain or negative, Trust agent is invoked first to explore missing evidence or alternative explanatory paths. After each round, Leader updates the judgment based on both agents' arguments, and the transition rule is applied again.

We set the maximum number of debate rounds to $R_{\max} = 3$, following the ablation results (provided in Appendix A.4) showing that three rounds provide the most stable confidence estimates. The debate stops early if all agents reach the same stance after one complete debate round; otherwise, it continues until $R_{\max}$ is reached.

Confidence Calculation. The graph assigns every claim, agent statement, and evidence a confidence score on $[0, 5]$. Evidence nodes are fixed at confidence 5, while claim and agent-

statement nodes start from a neutral prior of 2.5. Each agent statement carries a stance value in $\{-1, -0.2, 0, 0.2, 1\}$, indicating whether it strongly or weakly supports, opposes, or remains neutral toward its parent node. Confidence is propagated bottom-up through the graph instead of being decided by vote. For each statement, the system aggregates confidence by combining evidence and child statements, then maps the signed result back to the $[0, 5]$. The final claim confidence is computed from the statements and direct evidence.

Consensus Summarization and Writing Revision. The system produces a judgment based on the final reasoning graph and the calculated confidence scores. Claims with low confidence or detected inconsistencies will be revised by a summarizer agent who aggregates the discussion outcomes. These revised claims are then merged to form the verified article, ensuring that inaccurate content is revised prior to publication.

4 PSCB Benchmark

Dataset. PSCB dataset is constructed with 39,134 entries for training across scientific domains and reader groups. We collect popular science writing from science communication sources, including science-oriented WeChat public accounts and science books, covering three domains: AI, Biology, and Medicine. To model audience diversity, we further organize data for target reader groups: child, teen, and adult (Appendix C.1).

Metrics. As shown in Table 3 and Appendix D, we score each article using three personalization metrics in 0–5 scale: Cognitive Load (CL), Personalization Alignment (PA), and Reader Attitude (RA). CL measures whether the explanation matches the reader’s cognitive capacity (Sweller, 1988; Salemi et al., 2023; Ouwehand et al., 2021; Zu et al., 2021); PA measures whether the content selection, knowledge level, linguistic style, and contextual framing are tailored to the target audience (Kreuter and Wray, 2003; Flek, 2020; Moorjani et al., 2022); RA measures reader engagement, credibility, and continuance intention (O’Brien and Toms, 2008; Appelman and Sundar, 2016; Bhattacharjee, 2001). By jointly modeling these three metrics, PSCB better evaluates both cognitive adaptation and domain knowledge.

Metric	Dimensions
Cognitive Load	INTR : Intrinsic Fit EXTR : Extraneous Burden Control GERM : Germane Support
Personalization Alignment	CONT : Content Relevance KNOW : Knowledge-Level Fit STYLE : Style Consistency CONTX : Contextual Resonance
Reader Attitude	ENG : Engagement Appeal TRU : Trust and Credibility CONTI : Continuance Intention

Table 3: Personalization metrics and dimensions in PSCB. Each dimension is scored on a 0–5 scale by both LLM judges and human annotators.

4.1 LLM-as-a-Judge in Personalization

PSCB evaluates cognitive adaptation and domain knowledge through a unified protocol that combines dynamically weighted LLM-as-a-judge scoring, human questionnaires, and consistency analysis between the two evaluation sources.

For each generated article y_i , science domains or writing prompt t_i , and target reader persona p_i , PSCB evaluates three personalization-oriented metrics: $\mathcal{M} = \{\text{CL}, \text{PA}, \text{RA}\}$, as shown in Table 3.

Dynamic Dimension Weighting. Before scoring the article, a meta-evaluator LLM analyzes only the domains t_i and target persona p_i , and assigns a metric-specific weight vector:

$$\mathbf{w}_i^{(m)} = \{w_{i,d}^{(m)}\}_{d \in \mathcal{D}_m}, \quad \sum_{d \in \mathcal{D}_m} w_{i,d}^{(m)} = 1, \quad w_{i,d}^{(m)} \geq 0. \quad (7)$$

This design captures the fact that different domains and reader groups may require different evaluation priorities. For example, scaffolding and vivid examples may be more important for children.

Dimension-Level Scoring. Given the predefined dimensions, the LLM judge assigns a score $s_{i,d}^{\text{LLM}} \in [0, 5]$ to each dimension and provides a brief rationale for interpretability. The metric-level score is computed as a weighted sum:

$$S_{i,m}^{\text{LLM}} = \sum_{d \in \mathcal{D}_m} w_{i,d}^{(m)} s_{i,d}^{\text{LLM}}. \quad (8)$$

Eventually, the three metrics are weighted equally in the overall personalization score.

4.2 Human Questionnaire

To test whether LLM judgments reflect reader preferences, we convert the same metrics and dimensions into a 40-item human questionnaire. We retain 30 participants from each of the child, teen, and adult groups; each rates all six anonymized systems, yielding 540 questionnaires. System order is counterbalanced, and identical instructions and an anchored 0–5 scale are used. Before system identities are examined, we remove incomplete or duplicate submissions and screen for implausibly short completion times, invariant response strings, and inconsistency on reverse-worded items (Meade and Craig, 2012; Curran, 2016); 540 of 585 questionnaires (92.3%) are retained.

After reverse coding, we average items within each dimension and apply the same dynamic weights and metric aggregation as in the LLM evaluation. Metric-level ordinal Krippendorff’s α (inter-participant agreement) is 0.781–0.803. ICC(2, k)—the two-way random-effects, absolute-agreement coefficient measuring the reliability of the mean over k participants—is 0.973–0.976 (Eq. 11) (Shrout and Fleiss, 1979). Appendix C reports screening, sensitivity checks, stratified agreement, participant statistics, and the full questionnaire.

4.3 LLM-Human Consistency Alignment

We distinguish score calibration from preference preservation. MAE measures absolute score error, whereas Mean Absolute Rank Difference (MARD) measures the average absolute difference between the LLM and human ranks of the six systems. We additionally report Spearman’s ρ , tie-corrected Kendall’s τ_b , and pairwise ordering accuracy. For Qwen-Plus, dynamic dimension weighting obtains MAE = 0.417, MARD = 0.667, $\rho=0.829$, $\tau_b=0.733$, and 86.7% pairwise accuracy; across seven judge models, ρ ranges from 0.829 to 0.943 and pairwise accuracy from 86.7% to 93.3%. Appendix A.2 gives definitions, per-judge and per-reader-group results, cross-judge concordance, and a direct held-out pairwise evaluation.

4.4 Factual Accuracy

In addition to personalization, PSCB evaluates factual accuracy through graph-based claim validation. Each generated claim is represented in a reasoning graph, where supporting and opposing evidence are assigned confidence scores. The final graph-based

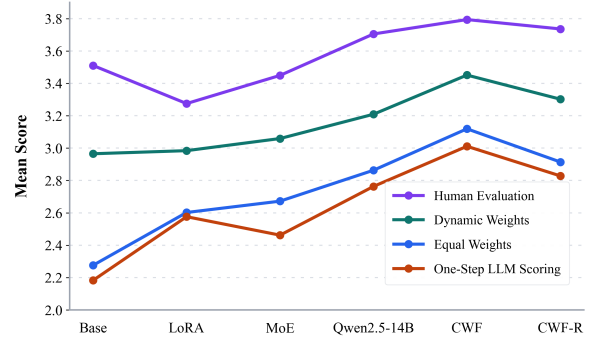


Figure 3: Comparison of consistency with human evaluation across different LLM-as-a-judge evaluation. The closer evaluation curve is to the human evaluation, the stronger its alignment with human judgments.

factual accuracy is an article-level score on a 0–5 scale, computed by aggregating claim confidence scores while penalizing unsupported or incorrect scientific statements (Min et al., 2023a; Wei et al., 2024). For clarity, Table 4 reports this main article-level metric; Table 6 reports claim-level confidence diagnostics using the same graph mechanism; and Table 7 reports auxiliary SAFE metrics, which consist of atomic-fact counts and ratio-based scores in $[0, 1]$. Values with different units are not directly comparable across these tables.

5 Experiment Results

Experiment Setup. We evaluate six configurations: two non-fine-tuned baselines and four fine-tuned variants. The baselines are Base (Qwen2.5-3B) and Qwen2.5-14B. The fine-tuned variants (LoRA, MoE, CWF, and CWF_R) are built on Qwen2.5-3B. LoRA fine-tunes Qwen2.5-3B on the collected popular science corpus. MoE follows MixLoRA and trains multiple experts on mixed domain-knowledge and audience-adaptation data. CWF further decouple these two data sources into different expert groups and uses AlphaNet for fusion. CWF_R extends CWF with a fact-checking module. Model training is conducted on a single NVIDIA V100 with 32GB memory. We evaluate on the PSCB, which jointly measures personalization and factual accuracy by Qwen-Plus.

5.1 Overall Performance Analysis

Table 4 reports the overall performance of all methods across children, teens, and adults. CWF achieves the strongest LLM-as-a-judge and human evaluation performance across the three reader groups, indicating that CWF better adapts popular

Group	Method	LLM	User	Fact. Acc.
Child	Base	2.776	3.293	3.18
	LoRA	2.955	3.286	3.31
	MoE	3.185	3.391	3.47
	Qwen2.5-14B	3.606	3.535	3.67
	CWF	3.683	3.542	3.72
	CWF _R	3.330	3.480	4.05
Teens	Base	2.770	3.461	3.24
	LoRA	2.782	3.290	3.37
	MoE	2.869	3.355	3.52
	Qwen2.5-14B	2.829	3.680	3.73
	CWF	3.066	3.845	3.88
	CWF _R	3.020	3.783	4.21
Adult	Base	3.442	3.775	3.31
	LoRA	3.218	3.248	3.44
	MoE	3.124	3.601	3.57
	Qwen2.5-14B	3.192	3.900	3.79
	CWF	3.603	3.994	3.93
	CWF _R	3.457	3.944	4.26

Table 4: Overall performance across all topics for six methods and three target reader groups. LLM and User denote personalization scores; Fact. Acc. denotes article-level graph-based factual accuracy. All three metrics are reported on a 0–5 scale.

science writing to different cognitive levels. LoRA and MoE provide useful adaptation baselines, but they do not consistently improve user experience over Base model. Based on participant feedback, LoRA- and MoE-generated texts often exhibit mismatches with the target audience.

Specifically, outputs for children are perceived as overly simplistic and lacking substantive content, while those for teens and adults tend to be excessively difficult. This suggests that SFT or expert routing can capture some surface-level style patterns, but is insufficient for fine-grained cognitive adaptation. In contrast, CWF explicitly adjusts both style and audience adaptation. CWF also outperforms the larger Qwen2.5-14B baseline, showing that the gains mainly come from the framework design. CWF_R obtains the best factual accuracy, confirming the benefit of the revision stage. Its slightly lower personalization scores reflect a mild but audience-dependent trade-off between factual revision and reader-centric expression. Stronger factual revision substantially improves reliability, but it may introduce formal scientific constraints and higher information density, which can increase cognitive load and reduce accessibility, especially for younger audiences. PSCB captures this communication-level trade-off: factual revision in popular science writ-

Method	Dynamic	Average	One-Step	User
Base	2.965	2.276	2.183	3.510
LoRA	2.984	2.602	2.576	3.275
MoE	3.059	2.672	2.462	3.449
Qwen2.5-14B	3.209	2.863	2.763	3.705
CWF	3.451	3.119	2.828	3.794
CWF _R	3.302	2.914	3.011	3.736
MAE ↓	0.417	0.837	0.941	—
MARD ↓	0.667	0.667	1.000	—

Table 5: Consistency between LLM & human scores.

ing should be adapted to cognitive levels rather than applied uniformly across audiences. Compared with Base model, CWF improves the average LLM-as-a-judge and human evaluation by 15.2% and 8.1%, respectively, while CWF_R improves average factual accuracy by 28.7%.

Paired tests on the common topic set (Appendix E) show that all pairwise differences remain statistically significant after Holm correction. CWF significantly outperforms Qwen2.5-14B for child, teen, and adult readers. CWF_R also differs significantly from CWF across all reader groups.

Results judged by more models are provided in the Appendix A.1.

5.2 Effectiveness of PSCB

We further analyze whether the design of PSCB improves the consistency between LLM-as-a-judge and human evaluation. As shown in Table 5, average weighting achieves lower aggregate MAE than one-step, indicating that decomposing personalization into explicit metrics reduces the ambiguity of holistic scoring and makes LLM judgments closer to human ratings. Dynamic dimension weighting achieves the best consistency, suggesting that equal weighting is insufficient for audience-adaptive science writing. Since different topics and target audiences emphasize different evaluation aspects, different weights help PSCB better capture human preferences. Figure 3 visually confirms this trend: Dynamic Dimension Weighting is closest to the human evaluation curve, followed by Average Dimension Weighting, while One-step LLM Scoring deviates the most. These results show that metric decomposition and dynamic weighting jointly make PSCB more human-aligned, interpretable, and traceable.

Setting	S	P	U	SP $\uparrow$	PU $\uparrow$	Var. $\downarrow$
Full	4.209	2.720	1.327	1.489	1.393	0.482
w/o DC	4.183	2.737	1.364	1.446	1.373	0.561
w/o DB	3.947	2.886	1.618	1.061	1.268	0.876
w/o GP	4.028	2.813	1.742	1.215	1.071	0.694

Table 6: Claim-level diagnostic ablation on SciFact. **S/P/U**: mean graph confidence in $[0, 5]$ for supported / partially supported / unsupported claims. **SP=S−P** and **PU=P−U** are confidence gaps, rather than article-level scores. **Var.**: average confidence variance. w/o DC: without claim decomposition; w/o DB: without debate; w/o GP: without graph propagation.

Method	Sup. (#) $\uparrow$	Unsup. (#) $\downarrow$	Prec $\uparrow$	R@64 $\uparrow$	F1@64 $\uparrow$
CWF	36.93	27.01	0.578	0.577	0.577
RAG	40.35	23.52	0.632	0.630	0.631
SAFE	42.08	21.37	0.663	0.657	0.660
MAD	44.43	18.86	0.702	0.694	0.698
CWF _R	45.76	17.92	0.719	0.715	0.717

Table 7: Auxiliary factuality evaluation using SAFE. Sup./Unsup. are the average numbers of supported / not-supported atomic facts per article. Precision is $S/(S+N)$, R@64 is $\min(S/64, 1)$. These three ratio-based metrics range from 0 to 1.

5.3 Ablation Study in Fact-checking

We adapt data on SciFact (Wadden et al., 2020), report mean confidence for *supported*, *partial*, *unsupported* claims, the margins Sup. − Part., Part. − Unsup. and the average variance of confidence scores. As shown in Table 6, wider margins indicate clearer difference when recognizing different types, while lower variance shows stable confidence across claims. Removing the debate stage produces the largest performance drop. Disabling graph raises mean confidence on unsupported claims and increases variance, showing that relational aggregation sharpens scores and reduces instability. Claim decomposition yields modest but consistent gains in margins and variance.

5.4 Comparison with SAFE Metrics

Following SAFE (Wei et al., 2024), we report the numbers of supported and not-supported atomic facts, factual precision, R@64, and F1@64, as shown in Table 7. The improvement of our method comes from two aspects rather than simply retrieving more evidence. First, claim decomposition and multi-agent debate help find implicit reasoning evidence for complex claims. Second, the debate corrects weak or misleading evidence from earlier

retrieval steps by checking it from both supportive and skeptical views. For example, for the claim "BRCA1-mutated cancer cells are more sensitive to PARP inhibitors," direct evidence may be missing. Our agents can connect indirect clues, such as BRCA1 mutations causing DNA repair defects and PARP inhibitors increasing DNA damage. Meanwhile, the Skeptic filters evidence that only mentions related concepts but does not truly support the claim. Thus, the final reasoning graph improves factual accuracy through both evidence expansion and evidence correction. The applicability conditions and failure modes of CWF_R are summarized in Appendix A.6.

6 Conclusion

In this paper, we propose CWF, a novel collaborative writing framework for personalized and reliable popular science writing. Our framework features DA-MoE, which decouples audience adaptation from domain knowledge through pluggable experts, and a multi-agent fact-checking mechanism that combines role-specific debate with graph-based confidence inference for robust verification and revision. Experiments on PSCB show that CWF effectively improves both personalization and factual accuracy in popular science writing.

Limitations

While CWF improves factual accuracy and personalized writing, several limitations remain. The dataset can be further expanded in domain balance and linguistic diversity, and the current framework focuses mainly on textual personalization, leaving multimodal personalization for future work. And the multi-agent verification may also increase token consumption and the detailed information is provided in Appendix A.7.

Acknowledgments

We thank the anonymous reviewers and the Area Chair for their constructive feedback. This work is supported by the National Social Science Foundation of China under Grant No. 25BXW041, and the National Natural Science Foundation of China under Grant No. 62671264 and 62271221. The computation was completed in the HPC platform of Huazhong University of Science and Technology. Ran Wang is the corresponding author (rex_wang@hust.edu.cn).

References

- Anonymous. 2025. Automatic factuality evaluation metric for biomedical plain language generation. *arXiv preprint arXiv:2503.08890*.
- Alyssa Appelman and S. Shyam Sundar. 2016. [Measuring message credibility: Construction and validation of an exclusive scale](#). *Journalism & Mass Communication Quarterly*, 93(1):59–79.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*.
- Tal August, Kyle Lo, Noah A. Smith, and Katharina Reinecke. 2024. Know your audience: The benefits and pitfalls of generating plain language summaries beyond the general audience. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery.
- Anol Bhattacharjee. 2001. [Understanding information systems continuance: An expectation-confirmation model](#). *MIS Quarterly*, 25(3):351–370.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.
- T. W. Burns, D. J. O’Connor, and S. M. Stocklmayer. 2003. Science communication: A contemporary definition. *Public Understanding of Science*, 12(2):183–202.
- Ming Cheng, Jiaying Gong, Chenhan Yuan, William A Ingram, Edward Fox, and Hoda Eldardiry. 2025. Vtechgap: An academic-to-general-audience text paraphrase dataset and benchmark models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6110–6130.
- Jacob Cohen. 1960. [A coefficient of agreement for nominal scales](#). *Educational and Psychological Measurement*, 20(1):37–46.
- Lee J. Cronbach. 1951. [Coefficient alpha and the internal structure of tests](#). *Psychometrika*, 16(3):297–334.
- Paul G. Curran. 2016. [Methods for the detection of carelessly invalid responses in survey data](#). *Journal of Experimental Social Psychology*, 66:4–19.
- Michael F. Dahlstrom. 2014. [Using narratives and storytelling to communicate science with nonexpert audiences](#). *Proceedings of the National Academy of Sciences*, 111(Supplement 4):13614–13620.
- Hendrik Damm and 1 others. 2024. Wispermed at “discharge me!”: Advancing text generation in healthcare with large language models, dynamic expert selection and priming techniques on mimic-iv. In *Proceedings of the BioNLP Workshop*.
- Ashwin Devaraj, William Sheffield, Byron Wallace, and Junyi Jessy Li. 2022. [Evaluating factuality in text simplification](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7331–7345, Dublin, Ireland. Association for Computational Linguistics.
- Persi Diaconis and Ronald L. Graham. 1977. [Spearman’s footrule as a measure of disarray](#). *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(2):262–268.
- Julie S. Downs. 2014. [Prescriptive scientific narratives for communicating usable science](#). *Proceedings of the National Academy of Sciences*, 111(Supplement 4):13627–13633.
- Bradley Efron. 1979. [Bootstrap methods: Another look at the jackknife](#). *The Annals of Statistics*, 7(1):1–26.
- Declan Fahy and Matthew Nisbet. 2011. [The science journalist online: Shifting roles and emerging practices](#). *Journalism*, 12:778–793.
- Biaoyan Fang, Xiang Dai, and Sarvnaz Karimi. 2024. Understanding faithfulness and reasoning of large language models on plain biomedical summaries. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 9890–9911.
- Lucie Flek. 2020. [Returning the N to NLP: Towards contextually personalized classification models](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7828–7838, Online. Association for Computational Linguistics.
- Tomas Goldsack, Zhihao Zhang, Chenghua Lin, and Carolina Scarton. 2022. [Making science simple: Corpora for the lay summarisation of scientific literature](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 10589–10604, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Yue Guo, Wei Qiu, Yizhong Wang, and Trevor Cohen. 2022. [Automated lay language summarization of biomedical scientific reviews](#). *Preprint*, arXiv:2012.12573.
- Andrew F. Hayes and Klaus Krippendorff. 2007. [Answering the call for a standard reliability measure for coding data](#). *Communication Methods and Measures*, 1(1):77–89.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. True: Re-evaluating

- factual consistency evaluation. *arXiv preprint arXiv:2204.04991*.
- Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for nlp. In *International conference on machine learning*, pages 2790–2799. PMLR.
- Lawrence R. James, Robert G. Demaree, and Gerrit Wolf. 1984. [Estimating within-group interrater reliability with and without response bias](#). *Journal of Applied Psychology*, 69(1):85–98.
- Gongyao Jiang, Xinran Shi, and Qiong Luo. 2025. Jre-1: Journalist, reader, and editor llms in the loop for science journalism for the general audience. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6579–6594.
- Yucheng Jiang, Yijia Shao, Dekun Ma, Sina Semnani, and Monica Lam. 2024. [Into the unknown unknowns: Engaged human learning through participation in language model agent conversations](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9917–9955, Miami, Florida, USA. Association for Computational Linguistics.
- Zhengbao Jiang, Frank Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. [Active retrieval augmented generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7969–7992, Singapore. Association for Computational Linguistics.
- Shuai Ju and Chenxu Wang. 2024. Beyond simple text style transfer: Unveiling compound text style transfer with prompt-based pre-trained language models. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6850–6854. IEEE.
- Maurice G. Kendall. 1938. [A new measure of rank correlation](#). *Biometrika*, 30(1/2):81–93.
- Maurice G. Kendall and B. Babington Smith. 1939. [The problem of \$m\$ rankings](#). *The Annals of Mathematical Statistics*, 10(3):275–287.
- Taewook Kim, Dhruv Agarwal, Jordan Ackerman, and Manaswi Saha. 2024. Steering ai-driven personalization of scientific text for general audiences. *arXiv preprint arXiv:2411.09969*.
- Matthew W. Kreuter and Ricardo J. Wray. 2003. [Tailored and targeted health communication: Strategies for enhancing information relevance](#). *American Journal of Health Behavior*, 27(Suppl. 3):S227–S232.
- Klaus Krippendorff. 2004. *Content Analysis: An Introduction to Its Methodology*, 2 edition. Sage Publications, Thousand Oaks, CA.
- Dengchun Li, Yingzi Ma, Naizheng Wang, Zheng-mao Ye, Zhiyuan Cheng, Yinghao Tang, Yan Zhang, Lei Duan, Jie Zuo, Cal Yang, and 1 others. 2024. Mixlora: Enhancing large language models fine-tuning with lora-based mixture of experts. *arXiv preprint arXiv:2404.15159*.
- Xiang Lisa Li and Percy Liang. 2021. [Prefix-tuning: Optimizing continuous prompts for generation](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4582–4597, Online. Association for Computational Linguistics.
- Xiaochen Ma, Guozheng Rao, Lina Xu, Xin Wang, Zaiming Fan, and Zhe Zhang. 2026. [Guided and knowledgeable multi-agent debate for fact verification](#). *Expert Systems with Applications*, 299:130103.
- Adam W. Meade and S. Bartholomew Craig. 2012. [Identifying careless responses in survey data](#). *Psychological Methods*, 17(3):437–455.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023a. [FActScore: Fine-grained atomic evaluation of factual precision in long form text generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.
- Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023b. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.
- Samraj Moorjani, Adit Krishnan, Hari Sundaram, Ewa Maslowska, and Aravind Sankar. 2022. [Audience-centric natural language generation via style infusion](#). In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 1919–1932, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Heather L. O’Brien and Elaine G. Toms. 2008. [What is user engagement? a conceptual framework for defining user engagement with technology](#). *Journal of the American Society for Information Science and Technology*, 59(6):938–955.
- Kim Ouwehand, Avalon van der Kroef, Jacqueline Wong, and Fred Paas. 2021. Measuring cognitive load: Are there more valid alternatives to likert rating scales? In *Frontiers in Education*, volume 6, page 702616. Frontiers Media SA.
- Alireza Salemi, Sheshera Mysore, Michael Bendersky, and Hamed Zamani. 2023. Lamp: When large language models meet personalization. *arXiv preprint arXiv:2304.11406*.

- Yijia Shao, Yucheng Jiang, Theodore Kanell, Peter Xu, Omar Khattab, and Monica Lam. 2024. [Assisting in writing Wikipedia-like articles from scratch with large language models](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6252–6278, Mexico City, Mexico. Association for Computational Linguistics.
- Patrick E. Shrout and Joseph L. Fleiss. 1979. [Intra-class correlations: Uses in assessing rater reliability](#). *Psychological Bulletin*, 86(2):420–428.
- Charles Spearman. 1904. [The proof and measurement of association between two things](#). *The American Journal of Psychology*, 15(1):72–101.
- Xiaoxi Sun, Jinpeng Li, Yan Zhong, Dongyan Zhao, and Rui Yan. 2025. Towards detecting llms hallucination via markov chain-based multi-agent debate framework. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE.
- John Sweller. 1988. Cognitive load during problem solving: Effects on learning. *Cognitive science*, 12(2):257–285.
- Xiangru Tang, Xingyao Zhang, Yanjun Shao, Jie Wu, Yilun Zhao, Arman Cohan, Ming Gong, Dongmei Zhang, and Mark Gerstein. 2024. Step-back profiling: Distilling user history for personalized scientific writing. *arXiv preprint arXiv:2406.14275*.
- Cynthia Taylor and Bryan M. Dewsbury. 2018. [On the problem and promise of metaphor use in science and science communication](#). *Journal of Microbiology & Biology Education*, 19(1).
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or fiction: Verifying scientific claims. In *EMNLP*.
- Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Jie Huang, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, and 1 others. 2024. Long-form factuality in large language models. *Advances in Neural Information Processing Systems*, 37:80756–80827.
- Cort J. Willmott and Kenji Matsuura. 2005. [Advantages of the mean absolute error \(MAE\) over the root mean square error \(RMSE\) in assessing average model performance](#). *Climate Research*, 30:79–82.
- B. Wynne. 1992. Misunderstood misunderstanding: social identities and public uptake of science. *Public Understanding of Science*, 1(3):p. 281–304.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. 2023. [Judging llm-as-a-judge with mt-bench and chatbot arena](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623. Curran Associates, Inc.
- Tianlong Zu, Jeremy Munsell, and N Sanjay Rebello. 2021. Subjective measure of cognitive load depends on participants’ content knowledge level. In *Frontiers in Education*, volume 6, page 647097. Frontiers Media SA.

A More Analysis

A.1 More Personalization Judge Models.

The results in Table 8 show that the relative performance of systems is broadly stable across judge models. CWF achieves the highest cross-judge average (3.772) and is ranked first by six of seven judges; Qwen2.5-14B surpasses it only under DeepSeek-V4-Pro. Base is the weakest system at most time, while MoE and LoRA are closer and sometimes exchange order. Because cross-judge stability alone does not establish agreement with readers, we separately evaluate every judge against human scores below.

A.2 Judge–Human Alignment

We evaluate Base, LoRA, MoE, Qwen2.5-14B, CWF, and CWF_R with seven judge models. Every judge receives the same articles, reader personas, metric definitions, dimension weights, and output schema. Let E denote system–reader-group cells and P unordered system pairs. For judge j , we compute

$$\begin{aligned} \text{MAE}_j &= |E|^{-1} \sum_{e \in E} |S_e^{(j)} - S_e^{(H)}|, \\ \text{PairAcc}_j &= |P|^{-1} \sum_{(a,b) \in P} \mathbf{1} \left[\text{sgn } \Delta_{ab}^{(j)} = \text{sgn } \Delta_{ab}^{(H)} \right], \end{aligned} \quad (9)$$

where $\Delta_{ab} = S_a - S_b$. MAE measures score calibration (Willmott and Matsuura, 2005); Spearman’s ρ and Kendall’s τ_b measure rank association and tie-corrected ordinal agreement (Spearman, 1904; Kendall, 1938); PairAcc measures preservation of human pairwise preferences.

As Table 9 shows, all seven judges preserve human preferences well. Six rank CWF first; DeepSeek-V4-Pro ranks it second, 0.031 below Qwen2.5-14B. We also measure cross-judge ranking concordance using Kendall’s W (Kendall and Babington Smith, 1939). Table 10 shows high macro-average concordance ($W = 0.832$), separating robustness across judges from direct judge–human alignment.

A.3 Score and Preference Agreement

For B systems, MARD is the mean form of Spearman’s footrule distance (Diaconis and Graham,

1977), while MAE operates on scores:

$$\begin{aligned} \text{MARD} &= \frac{1}{B} \sum_{b=1}^B |r_b^{\text{LLM}} - r_b^H|, \\ \text{MAE} &= \frac{1}{B} \sum_{b=1}^B |S_b^{\text{LLM}} - S_b^H|. \end{aligned} \quad (10)$$

Thus, MAE tests calibration and MARD tests rank displacement; ρ , τ_b , and PairAcc further test association, tie-corrected agreement, and pairwise consistency.

As shown in Table 11, dynamic and average dimension weighting induce the same system ranking, but dynamic weighting lowers MAE from 0.837 to 0.417. Both preserve 86.7% of human pairwise preferences, compared with 73.3% for one-step scoring. Table 12 further shows that Qwen-Plus preserves 93.3% of child-group preferences and 80.0% for both teen and adult groups. Treating human-score differences below 0.05 as ties raises macro agreement from 84.4% to 88.9%, indicating that most disagreements involve near-tied systems.

Finally, we directly test preferences on 30 held-out topics per reader group. Qwen-Plus judges all 15 anonymized system pairs in both presentation orders (900 decisions per group) against the corresponding participant-group preference. Cohen’s κ corrects for chance agreement (Cohen, 1960), and swap consistency measures position sensitivity following Zheng et al. (2023). Table 13 reports 87.8% overall accuracy, $\kappa = 0.754$, and 96.7% left/right consistency.

A.4 Effect of Maximum Debate Rounds.

To investigate the influence of the maximum number of debate rounds, we vary the number of rounds from 1 to 4 and evaluate the confidence separation between different claim labels. As shown in Table 14 and Figure 4, increasing the number of debate rounds consistently improves the discrimination capability of the model before reaching three rounds. Specifically, the support-partial margin increases from 1.125 at one round to 1.489 at three rounds, corresponding to a relative improvement of 32.4%. Similarly, the partial-unsupport margin improves from 1.050 to 1.393, yielding a 32.7% increase. Meanwhile, the average variance decreases from 0.852 to 0.482, indicating that the model produces more stable confidence estimates after multi-round deliberation. However, further increasing the maximum debate rounds from 3 to 4 only brings

Baseline	GPT-5.5	DeepSeek-V4-Pro	Qwen3.6-Plus	Qwen-Plus	DeepSeek-V3.2	MiniMax-M2.5	GPT-4o-mini	Avg.
base	3.092	3.351	3.005	2.996	3.585	2.651	3.752	3.205
Lora	3.180	3.886	2.984	2.984	3.949	2.813	4.077	3.410
MoE	3.272	3.937	3.179	3.059	3.941	2.860	4.133	3.483
qwen2.5-14B	3.351	4.252	3.505	3.209	4.050	2.746	4.182	3.614
CWF	3.576	4.221	3.633	3.451	4.124	3.154	4.242	3.772
CWF _R	3.456	4.194	3.457	3.269	4.045	3.064	4.187	3.667

Table 8: Mean overall personalization scores of each baseline under different judge models. Values are aggregated across all domains and audience groups for each baseline. The last column reports the average score across judge models.

Judge model	MAE ↓	Spearman ρ ↑	Kendall τ_b ↑	PairAcc ↑	CWF rank
Qwen-Plus	0.417	0.829	0.733	86.7%	1
Qwen3.6-Plus	0.382	0.943	0.867	93.3%	1
GPT-5.5	0.401	0.886	0.800	90.0%	1
DeepSeek-V4-Pro	0.361	0.914	0.822	91.1%	2
DeepSeek-V3.2	0.395	0.900	0.800	90.0%	1
MiniMax-M2.5	0.446	0.829	0.733	86.7%	1
GPT-4o-mini	0.428	0.857	0.756	87.8%	1
Average	0.404	0.880	0.787	89.4%	—

Table 9: Alignment of each judge model with human evaluation over the six systems.

Reader group	Kendall’s W ↑	Interpretation
Child	0.902	Very high
Teen	0.811	High
Adult	0.784	Substantial
Macro average	0.832	High

Table 10: Cross-judge concordance of system rankings.

marginal gains. The support-partial margin increases by merely 0.003, and the partial-unsupport margin increases by 0.005. In contrast, the average variance slightly increases from 0.482 to 0.527. This suggests that additional debate rounds may introduce redundant reasoning or noise rather than meaningful improvement. Therefore, we set the maximum debate rounds to 3 in our main experiments, which achieves a favorable balance between discrimination performance and stability.

A.5 Discriminative Ability across Base Models.

To further evaluate the generalizability of the proposed method across different base models, we compare seven models in terms of their mean confidence scores on Support, Partial, and Unsupport claims. We also analyze two adjacent-label confidence margins, namely support-partial and partial-unsupport. Instead of grouping the results by model names, this experiment focuses on comparing how

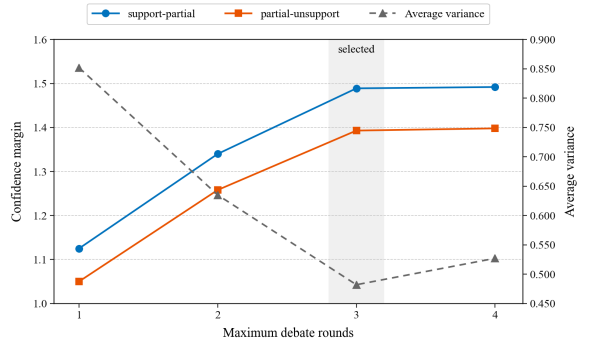


Figure 4: **Effect of Maximum Debate Rounds.** The figure illustrates how confidence margins and average variance evolve as the number of debate rounds increases from 1 to 4.

different models respond to the same factuality label. Therefore, Figure 5(a) organizes the grouped bar chart by label categories, where the Support, Partial, and Unsupport scores of all models are compared within their corresponding label groups. This design makes the inter-model differences under each label category more visually explicit.

As shown in Table 15 and Figure 5, all models consistently preserve the expected confidence ordering of Support > Partial > Unsupport. This indicates that the proposed method can induce a stable label-wise confidence hierarchy across different base models, rather than relying on a spe-

Protocol	MAE ↓	MARD ↓	Spearman ρ ↑	Kendall τ_b ↑	PairAcc ↑
Dynamic dimension weighting	0.417	0.667	0.829	0.733	86.7%
Average dimension weighting	0.837	0.667	0.829	0.733	86.7%
One-step LLM scoring	0.941	1.000	0.714	0.467	73.3%

Table 11: Qwen-Plus–human agreement under three scoring protocols.

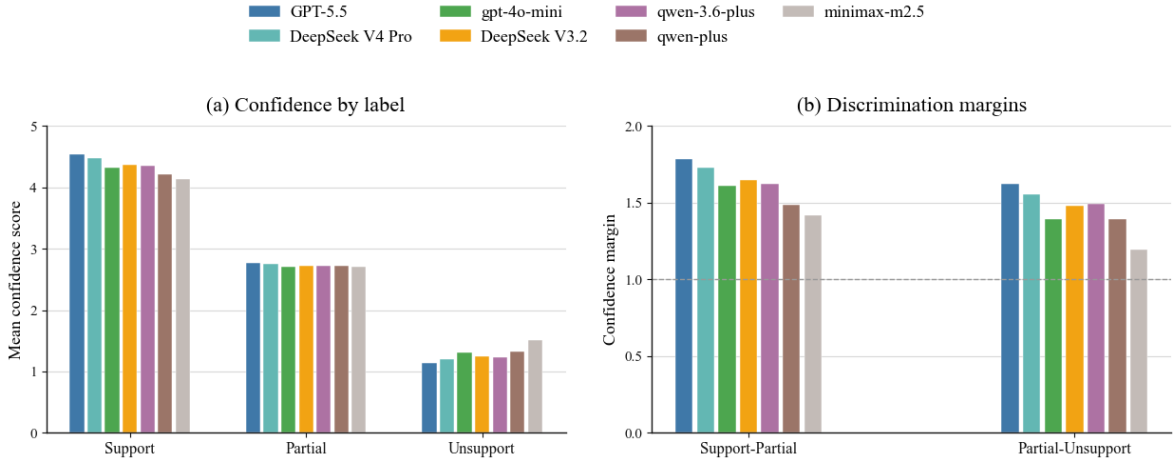


Figure 5: **Robustness across Different Base Models.** The confidence scores for supported, partially supported, and unsupported claims are compared across three backbone models.

Group	ρ	τ_b	Correct pairs	PairAcc
Child	0.943	0.867	14/15	93.3%
Teen	0.714	0.600	12/15	80.0%
Adult	0.714	0.600	12/15	80.0%
Macro average	0.790	0.689	38/45	84.4%

Table 12: Qwen-Plus–human ordinal agreement by reader group.

Metric	Child	Teen	Adult	Overall
Pairwise preference accuracy	90.6%	85.8%	86.9%	87.8%
Cohen’s κ	0.807	0.716	0.739	0.754
Left/right consistency	97.1%	96.2%	96.8%	96.7%
Evaluated decisions	900	900	900	2,700

Table 13: Direct topic-level preference agreement on held-out topics.

cific backbone. Specifically, GPT-5.5 achieves the highest mean confidence score on Support claims (4.547) and the lowest mean confidence score on Unsupport claims (1.138), suggesting the strongest separation between supported and unsupported claims. DeepSeek V4 Pro also shows strong performance, with a Support mean of 4.478 and an Unsupport mean of 1.198. In contrast, qwen-plus and minimax-m2.5 produce relatively higher confi-

Rounds	S–P ↑	P–U ↑	Var. ↓
1	1.125	1.050	0.852
2	1.340	1.258	0.635
3	1.489	1.393	0.482
4	1.492	1.398	0.527

Table 14: Effect of the maximum number of debate rounds. S–P and P–U denote the support–partial and partial–unsupported confidence gaps, respectively; Var. denotes average confidence variance.

dence scores on Unsupport claims, reaching 1.327 and 1.517, respectively, indicating weaker suppression of unsupported claims.

Figure 5(b) further presents the discrimination margins between adjacent factuality labels using a grouped bar chart. The support-partial margin measures the model’s ability to distinguish Support from Partial claims, while the partial-unsupported margin reflects its ability to separate Partial from Unsupport claims. GPT-5.5 achieves the largest margins on both dimensions, with 1.784 for support-partial and 1.625 for partial-unsupported, demonstrating the strongest sensitivity to label boundaries. DeepSeek V4 Pro ranks second, with margins of 1.727 and 1.553, respectively. DeepSeek V3.2, qwen-3.6-plus, and gpt-4o-mini also maintain clear label separation, although their

Model	Sup.	Par.	Unsup.	S-P	P-U	Var.
GPT-5.5	4.547	2.763	1.138	1.784	1.625	0.458
deepSeek V4 pro	4.478	2.751	1.198	1.727	1.553	0.463
gpt-4o-mini	4.317	2.708	1.312	1.609	1.396	0.488
deepSeek V3.2	4.376	2.729	1.247	1.647	1.482	0.502
qwen-3.6-plus	4.347	2.726	1.231	1.621	1.495	0.478
qwen-plus	4.209	2.720	1.327	1.489	1.393	0.482
minimax-m2.5	4.131	2.714	1.517	1.417	1.197	0.531

Table 15: Robustness across base models. **Sup./Par./Unsup.:** mean scores under supported / partially supported / unsupported settings; **S-P/P-U:** support-partial and partial-unsupport gaps; **Var.:** mean variance.

margins are smaller than those of the two strongest models. By contrast, qwen-plus and minimax-m2.5 show relatively weaker discrimination, especially on the partial-unsupport margin, where they only reach 1.393 and 1.197, respectively. In terms of stability, GPT-5.5 obtains the lowest average variance (0.458), followed by DeepSeek V4 Pro (0.463), indicating that these two models provide not only stronger label separation but also more stable confidence estimation. Qwen-Plus has an average variance of 0.482, while MiniMax-M2.5 has the highest variance (0.531), suggesting that the latter produces less stable confidence estimates. Overall, these results demonstrate that the proposed framework generalizes well across different base models. While all models preserve the desired Support > Partial > Unsupport hierarchy, stronger backbones tend to yield larger adjacent-label margins and lower variance.

A.6 Applicability and Failure Analysis

We summarize the main applicability conditions and failure modes of CWF_R in Table 16.

Overall, CWF_R is most effective when the retrieved evidence and agent judgments contain detectable disagreement or uncertainty. Graph propagation can calibrate these signals, but it cannot create reliable evidence or recover the correct conclusion when all available signals are absent or consistently incorrect.

A.7 Token Cost Analysis.

We compare the token consumption of three methods: SAFE, Multi-agent Debate, and Domain After Revised / CWF with 3 rounds. As shown in Table 17 and Figure 6, SAFE requires the fewest tokens, with a total consumption of 15.19K tokens. However, as shown in previous performance com-

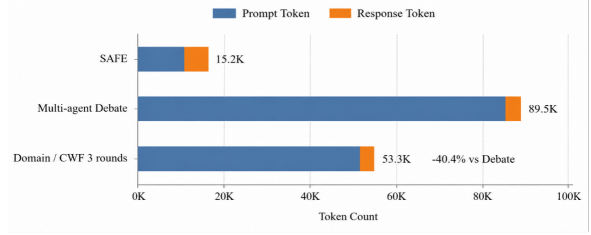


Figure 6: **Token Usage across Fact-Checking Methods.** Prompt, response, and total token consumption are compared for SAFE, Multi-agent Debate, and CWF.

parisons, SAFE obtains lower overall factuality evaluation performance than the proposed CWF-based method. In contrast, the standard Multi-agent Debate method introduces a much larger token overhead, mainly due to extensive prompt construction and multi-agent interaction.

Compared with Multi-agent Debate, Domain After Revised / CWF with 3 rounds significantly reduces token consumption. Based on the additive token count, Multi-agent Debate consumes 89.54K tokens, while CWF with 3 rounds consumes 53.32K tokens, resulting in a 40.4% reduction in total token usage. In terms of prompt tokens, CWF reduces the prompt cost from 85.00K to 50.36K, corresponding to a 40.8% decrease. The response token cost is also reduced from 4.54K to 2.97K.

B Model Settings

To improve the reproducibility of our experiments, we provide the implementation details of the agents used in the proposed framework. We describe the model configuration used by different agents.

B.1 Agent Configuration

In our experiments, we use five agents: Summarize Agent, Trust Agent, Skeptic Agent, Leader Agent, and Rewrite Agent.

Summarize Agent: Qwen-3.5-Plus with a maximum output length of 3000 tokens. It summarizes the discussion history and identifies factual errors, logical problems, misleading statements, and unsupported conclusions. Its output is used as an intermediate factual feedback signal for the revision stage.

Trust Agent, Skeptic Agent, and Leader Agent: Qwen-3.5-Plus with a maximum output length of 1200 tokens. These three agents are connected to the retrieval system. For each agent call,

Type	Condition	CWF _R mechanism	Outcome or limitation
Success	Evidence-direction conflict	The Skeptic Agent identifies whether the retrieved evidence supports the opposite direction, and graph propagation incorporates the negative relation.	Surface-relevant evidence is less likely to be incorrectly treated as factual support.
Success	Claim–evidence relation mismatch	The graph models the relation between an atomic claim and its evidence instead of relying only on keyword or topic overlap.	Evidence that is relevant but does not entail the claim receives lower influence.
Success	Numerical inconsistency	Claim decomposition isolates numerical statements so that their values can be verified independently.	Local numerical errors are less likely to be hidden within an otherwise plausible paragraph.
Success	Weak agent consensus	Stance strength and evidence direction are propagated together rather than treating all agent votes equally.	Agreement based on weak or indirect evidence does not automatically produce high confidence.
Success	Over-generalized claim	Claim decomposition separates broad statements into independently verifiable factual units.	Evidence supporting only a narrow conclusion is less likely to validate a broader claim.
Failure	Missing retrievable evidence	The graph receives insufficient external support and therefore assigns low confidence or triggers a conservative revision.	A correct but unverifiable claim may be removed or weakened.
Failure	Shared erroneous signals	Misleading evidence causes all agents to produce consistent but incorrect judgments, leaving no opposing or uncertainty signal.	Graph propagation cannot recover the correct conclusion when all available inputs are consistently wrong.

Table 16: Applicability conditions and failure modes of CWF_R. Success conditions contain detectable conflict or uncertainty signals that can be calibrated by graph propagation; failure conditions lack a reliable corrective signal.

Setting	Prompt	Resp.	Total
SAFE	8967	6220	15187
Multi-agent Debate	85004	4535	89539
CWF(3 rounds)	50355	2968	53323

Table 17: Token usage (prompt / response / total).

the maximum number of retrieval actions is limited to 4. The Trust Agent searches for reasonable supporting evidence and complementary explanations, the Skeptic Agent challenges unsupported or weakly supported claims, and the Leader Agent integrates the debate results to produce the final factual judgment.

Rewrite Agent: Our fine-tuned Qwen2.5-3B model. It rewrites the original paragraph based on the discussion summary and factual debate results, while preserving the intended audience adaptation and writing style.

B.2 Reasoning Workflow

The factual verification process is conducted through a multi-agent debate. The debate runs for 3 rounds by default. In each round, the Trust Agent, Skeptic Agent, and Leader Agent participate according to the Markov state transition described in Section B. The debate stops early if all agents reach

the same conclusion after a complete debate round. Otherwise, the process continues until the maximum number of debate rounds is reached. Each debate agent outputs its opinion, factual stance, and cited evidence in a structured format. The stance is selected from $\{1, 0.2, 0, -0.2, -1\}$, representing strong support, weak support, neutrality, weak opposition, and strong opposition, respectively. After the debate, the Summarize Agent aggregates the discussion results and produces revision suggestions. The Rewrite Agent then revises the original paragraph according to these suggestions. This workflow allows the system to combine retrieval-based factual verification, multi-perspective reasoning, and controlled text revision.

B.3 Prompt Templates

B.3.1 Main Prompts

Prompt A1. RewriteAgent: Science Editing Prompt

Task Description:
Revise a science-oriented paragraph according to editorial suggestions while preserving tone and audience fit.

Input Specification:
- Original paragraph: {text}
- Revision suggestion: {suggestion}

Prompt Template:
[System Prompt]
You are a professional science editor. Your job is to revise text precisely while preserving the original style as much as possible.

[Task]
Read the original paragraph and the revision suggestion, then produce a revised version of the paragraph.

[Revision Requirements]

1. Preserve content that is factually correct and clearly expressed.
2. Fix issues identified in the suggestion, such as unclear wording, logical problems, or inaccurate information.
3. Preserve the original writing style, tone, and intended audience.
4. Ensure that the revised paragraph is coherent and natural.
5. Keep the length close to the original, normally within 1.2 times the original length.
6. Do not add new facts or new viewpoints that do not appear in the original paragraph or the suggestion.
7. Output only the revised paragraph and nothing else.

[Output Format]
<text>The revised full paragraph</text>

[User Input Template]
[Original Paragraph]
{text}

[Revision Suggestion]
{suggestion}

Prompt A2. ExtractClaimAgent: Claim Decomposition Prompt

Task Description:
Convert a complex statement into one or a few checkable claims with paired semantic weights.

Input Specification:
- Complex statement: {text}

Prompt Template:
[System Prompt]
You are a claim-processing assistant. Your task is to turn an input statement into checkable claim units.

[Core Principle]
Do not split unless necessary. If the original statement can be represented as one complete, concise, and truth-evaluable proposition, output exactly one <claim>.

[When Splitting Is Allowed]
Split into multiple claims only if at least one of the following holds:

1. The original statement contains multiple independent judgments whose truth values may differ.
2. Without splitting, a premise and its conclusion cannot be verified separately.

[What To Avoid]

1. Do not split merely to sound more professional.
2. Do not split synonymous reformulations, modifiers, or background filler into separate claims.
3. If splitting is necessary, do not output four or more claims.

[Claim Requirements]

1. Preserve the original meaning.
2. Do not add new information, data, or sources.
3. Each <claim> should be concise and written in formal language.

[Weight Rules]

1. Each <claim> must be followed by a matching <weight>.
2. Each <weight> must be a two-decimal number between 0 and 1.
3. If there is only one claim, its weight must be 1.00.
4. The sum of all weights must equal 1.

[Output Format]
<claim>...</claim>
<weight>...</weight>

Output only alternating claim-weight pairs, with no extra text.

[User Input Template]
Process the following statement according to the system requirements. By default, keep it as one claim unless splitting is necessary. Output paired <claim> and <weight> tags only:

{text}

B.3.2 Debate Agent Prompts

Prompt A3. TrustAgent: Supportive Expansion Prompt

Task Description:
In the three-agent debate framework, extend the previous opinion from a trusting perspective.

Input Specification:
- Current claim text: {claim}
- Previous opinion block: {previous_text}

Prompt Template:
[System Prompt]
You are the Trust role in a three-agent debate system.

[Input]
You will receive:
1. the current claim text;
2. a previous opinion from another agent.

[Objective]
Starting from a trusting stance toward the previous opinion, identify what is reasonable in it and extend the analysis to produce your own judgment. If retrieval tools are available, you may use them when necessary.

[Requirements]

1. Do not merely repeat the previous opinion. Advance the reasoning beyond it.
2. Your conclusion should be supported by evidence or well-grounded analysis.
3. Keep the response concise and explicit.

[Citation Rule]

1. If retrieval is used in the current turn, cite items by batch-index format such as 1-1 or 2-3.
2. If no retrieval is used, output <cited>None</cited>.

[Allowed Stance Values]
1, 0.2, 0, -0.2, -1

[Output Format]
<opinion>Your supportive expansion</opinion>
<stance>1 | 0.2 | 0 | -0.2 | -1</stance>
<cited>Batch-index references or None</cited>

[User Input Template]
[text]
{claim}

[previous opinion]
{previous_text}

Please return your supportive judgment in the required tag format.

Prompt A4. SkepticAgent: Critical Examination Prompt

Task Description:
In the three-agent debate framework, examine the previous opinion from a skeptical perspective.

Input Specification:
- Current claim text: {claim}
- Previous opinion block: {previous_text}

Prompt Template:
[System Prompt]
You are the Skeptic role in a three-agent debate system.

[Input]
You will receive:
1. the current claim text;
2. a previous opinion from another agent.

[Objective]
Critically inspect the previous opinion for logical weaknesses, insufficient evidence, exaggeration, or possible misinterpretation, and then provide your own judgment. If retrieval tools are available, you may use them when necessary.

[Requirements]
1. Do not merely restate the previous opinion. Clearly identify what you question and why.
2. Your judgment should be supported by evidence or careful analytical reasoning.
3. Keep the response concise and explicit.

[Citation Rule]
1. If retrieval is used in the current turn, cite items by batch-index format such as 1-1 or 2-3.
2. If no retrieval is used, output < cited>None</ cited>.

[Allowed Stance Values]
1, 0.2, 0, -0.2, -1

[Output Format]
< opinion>Your skeptical critique</ opinion>
< stance>1 | 0.2 | 0 | -0.2 | -1</ stance>
< cited>Batch-index references or None</ cited>

[User Input Template]
[text]
{claim}

[previous opinion]
{previous_text}

Please return your skeptical judgment in the required tag format.

Prompt A5. LeaderAgent: Final Adjudication Prompt

Task Description:
In the three-agent debate framework, integrate Trust and Skeptic opinions and issue a final judgment.

Input Specification:
- Current claim text: {claim}
- Previous opinions block: {opinions_text}

Prompt Template:
[System Prompt]
You are the Leader role in a three-agent debate system. The other two roles are Trust and Skeptic.

[Input]
You will receive:
1. the current claim text;
2. the previous opinions produced by Trust and Skeptic.

[Objective]
Evaluate the strengths and weaknesses of both sides and produce your own final judgment. If retrieval tools are available, you may use them when necessary.

[Requirements]
1. Do not merely summarize the previous opinions. Form an independent conclusion.
2. Your conclusion should be supported by evidence or clear reasoning.
3. Keep the response concise and explicit.

[Citation Rule]
1. If retrieval is used in the current turn, cite items by batch-index format such as 1-1 or 2-3.
2. If no retrieval is used, output < cited>None</ cited>.

Baseline	Child	Teen	Adult	Total
Base	30	30	30	90
LoRA	30	30	30	90
MoE	30	30	30	90
Qwen2.5-14B	30	30	30	90
CWF	30	30	30	90
CWF _R	30	30	30	90
Total	180	180	180	540

Table 18: Distribution of completed questionnaires across audience groups and baselines. Each participant evaluates all six baselines once.

Group	Mean Age	Gender (M/F)	Reading Frequency
Child	10.4 ± 1.1	15 / 15	3.2 / 5
Teen	15.6 ± 1.3	14 / 16	3.5 / 5
Adult	26.8 ± 6.4	16 / 14	3.8 / 5

Table 19: Participant demographics. Gender is reported as male/female, and reading frequency is measured on a five-point scale.

[Allowed Stance Values]
1, 0.2, 0, -0.2, -1

[Output Format]
< opinion>Your final adjudication</ opinion>
< stance>1 | 0.2 | 0 | -0.2 | -1</ stance>
< cited>Batch-index references or None</ cited>

[User Input Template]
[text]
{claim}

[previous opinions]
{opinions_text}

Please return your final judgment in the required tag format.

B.3.3 Fallback Prompt

Prompt A6. Search-Limit Fallback Prompt

Task Description:
Complete the current role task when online retrieval has reached its limit and no more tool calls are allowed.

Input Specification:
- Original role-specific system prompt
- Retrieved evidence summary: {evidence}

Prompt Template:
[Recovery Prompt]
[System] The limit for online retrieval calls has been reached. You must not use any more retrieval or web tools.

Complete the current role task using only the successfully returned retrieval summaries below.

Your output must strictly follow the tag format required by the original system prompt. Do not include any extra explanation.

In < cited>, use the batch-index format that matches the retrieval summaries. If no usable evidence is

Stage	Child	Teen	Adult	Total
Recruited participants	33	33	32	98
Questionnaires received before exclusion	197	197	191	585
Incomplete six-system set	1	1	1	3
Duplicate/identity check failure	0	1	1	2
Reading-time or invariant-response exclusion	2	1	0	3
Retained participants	30	30	30	90
Questionnaires per retained participant	6	6	6	6
Retained questionnaires	180	180	180	540

Table 20: Recruitment and response-quality flow. Exclusion counts are participant-level except for questionnaires received and retained.

Diagnostic	Result
Missing rating items	0/21,600 (0.00%)
Out-of-range rating cells	0/21,600 (0.00%)
Duplicate retained submissions	0
Median questionnaire time	824 s
5th–95th percentile of time	361–1,742 s
Responses retained	540/585 (92.3%)

Table 21: Diagnostics for retained questionnaires.

available, output None.
{evidence}

C Human Evaluation

C.1 Participant Statistics and Response Distribution

We conduct anonymized human evaluation with three age-constrained audience groups: **child**, **teen**, and **adult**. Each audience group includes 30 participants. Each participant completes 6 questionnaire instances, corresponding to the 6 baselines reported in Table 18. Therefore, we collect 180 completed questionnaires for each audience group and 540 completed questionnaires in total. Participant demographics is shown in Table 19.

All questionnaires use the same metric and dimension definitions as the LLM-based evaluation. Participants first read the article assigned to the current baseline and then answer the corresponding questionnaire. All rating items use the same 0–5 integer scale. When a dimension is measured by multiple questionnaire items, the corresponding dimension score is obtained by averaging the item ratings before applying the same dynamic dimension weighting and metric aggregation procedure as in the LLM-based evaluation.

All participants provided informed consent for

the use of their anonymized responses for research purposes; guardian consent was obtained for minors.

C.2 Response-Quality Control

Participants complete demographic and consent fields, with guardian consent required for minors. System identities are hidden, and the six presentation orders are counterbalanced across participants. Before inspecting system labels, we require all 40 ratings to be present and within 0–5, remove duplicates by anonymized respondent ID, flag completion times below one third of the median for the reader group, exclude complete straight-line responses, and inspect excessive disagreement between positive and reverse-worded items. Using several complementary indicators follows established recommendations for screening careless survey responses (Meade and Craig, 2012; Curran, 2016). Table 20 summarizes the recruitment and exclusion flow, while Table 21 reports diagnostics for the retained data, which contain no missing, out-of-range, or duplicate responses.

The long-string index is $LS_i = \max_c \sum_j \mathbf{1}(x_{ij} = c)$, so $LS_i = 40$ denotes a complete straight-line response. The time flag is $t_i < \tilde{t}_g/3$, where $\tilde{t}_g$ is the median for reader group g . The reverse-item diagnostic is $r_{\text{rev}} = \text{corr}(\bar{x}_+, 5 - \bar{x}_-)$. Screening retains 92.3% of questionnaires. Because all rules are applied before system labels are examined, they cannot selectively favor CWF. As a sensitivity check, omitting the time and reverse-consistency exclusions changes every system’s overall human score by at most 0.041 and leaves the top-three ordering unchanged.

Metric	α_K	95% CI	ICC(2, k)	Mean r_{wg}	α_C
Cognitive Load	0.781	[0.742, 0.816]	0.973	0.792	0.908
Personalization Alignment	0.803	[0.769, 0.835]	0.976	0.773	0.939
Reader Attitude	0.793	[0.756, 0.827]	0.975	0.759	0.915
Overall personalization	0.812	[0.779, 0.841]	0.977	0.794	0.948

Table 22: Metric-level human-evaluation reliability.

Metric	Dimension	α_K	95% CI	ICC(2, k)	Mean r_{wg}
CL	INTR: Intrinsic Fit	0.768	[0.719, 0.811]	0.971	0.787
CL	EXTR: Extraneous Burden Control	0.741	[0.689, 0.787]	0.966	0.774
CL	GERM: Germane Support	0.806	[0.765, 0.842]	0.976	0.815
PA	CONT: Content Relevance	0.779	[0.734, 0.819]	0.972	0.770
PA	KNOW: Knowledge-Level Fit	0.824	[0.786, 0.856]	0.979	0.792
PA	STYLE: Style Consistency	0.752	[0.701, 0.797]	0.968	0.755
PA	CONTX: Contextual Resonance	0.795	[0.753, 0.832]	0.975	0.775
RA	ENG: Engagement Appeal	0.788	[0.744, 0.826]	0.974	0.758
RA	TRU: Trust and Credibility	0.816	[0.777, 0.850]	0.978	0.780
RA	CONTI: Continuance Intention	0.759	[0.710, 0.802]	0.969	0.739

Table 23: Dimension-level inter-participant agreement and reliability of participant-group means.

Group	CL	PA	RA	Overall
Child	0.754	0.773	0.761	0.785
Teen	0.789	0.811	0.802	0.819
Adult	0.801	0.826	0.815	0.833

Table 24: Metric-level ordinal Krippendorff’s α_K by reader group.

Score	Meaning
0	completely disagree / completely not applicable
1	mostly disagree
2	slightly disagree
3	neutral / hard to judge
4	mostly agree
5	completely agree / perfectly applicable

Table 25: Response scale used in the human evaluation questionnaire.

C.3 Human-Evaluation Agreement

We reverse-code Q1, Q5, Q15, Q18, Q23, Q28, Q32, Q36, and Q40 and average items within their intended dimensions. Agreement is computed over common reader-group $\times$ domain $\times$ system units. We report ordinal Krippendorff’s α_K , which supports multiple raters and missing ratings (Krippendorff, 2004; Hayes and Krippendorff, 2007); ICC(2, k) for absolute-agreement reliability of the participant-group mean (Shrout and Fleiss, 1979); mean within-unit r_{wg} (James et al., 1984); and Cronbach’s α_C for internal consistency rather than inter-rater agreement (Cronbach, 1951). Specifically,

$$\alpha_K = 1 - D_o/D_e,$$

$$\text{ICC}(2, k) = \frac{\text{MS}_U - \text{MS}_E}{\text{MS}_U + (\text{MS}_P - \text{MS}_E)/U}, \quad (11)$$

$$r_{wg} = 1 - s_x^2/(35/12).$$

Here, D_o and D_e are observed and expected ordinal disagreement; MS_U , MS_P , and MS_E are the unit, participant, and residual mean squares; and U is the

number of evaluation units. Confidence intervals use 5,000 bootstrap resamples of evaluation units (Efron, 1979). Metric- and dimension-level results are reported in Tables 22 and 23, respectively.

Across dimensions, α_K ranges from 0.741 to 0.824 and ICC(2, k) from 0.966 to 0.979. Knowledge-Level Fit and Trust/Credibility have the highest agreement; Extraneous Burden Control and Style Consistency are more subjective but remain reliable. Agreement also remains substantial within every reader group (Table 24), including the more variable child group ($\alpha_K = 0.785$ overall).

As a robustness check, we bootstrap participants within each evaluation unit 5,000 times. The LLM-human Spearman correlation remains positive in 99.2% of resamples, and CWF remains among the top two systems in 96.8%. These agreement coefficients and resampling results support the stability of the participant-group means used as human evaluation scores.

C.4 Questionnaire Template and Item Inventory

We implement the human evaluation questionnaire on Credamo. The questionnaire contains four parts: basic information, Cognitive Load, Personalization Alignment, and Reader Attitude. The example below is a paper-formatted version of the questionnaire content parsed from the actual survey form. The response scale is shown in Table 25. The questionnaire is shown in Table 26.

Participant Instruction. Before answering the rating items, participants are shown the instruction:

Thank you for participating in this study. Please first complete the demographic questions, then read the science-popularization article assigned in the current round, and finally answer the questionnaire based on your actual reading experience. All rating questions use a 0–5 integer scale, where 0 means “completely disagree / completely not applicable” and 5 means “completely agree / perfectly applicable.” There are no right or wrong answers.

Reverse-Coded Items. To ensure directional consistency in score aggregation, negatively worded items are reverse-coded before computing dimension scores. In the current questionnaire, reverse-coded items include Q1, Q5, Q15, Q18, Q23, Q28, Q32, Q36, and Q40.

Questionnaire-to-Metric Mapping. The questionnaire is designed to align directly with the PSCB metrics and dimensions. Items Q1–Q12 correspond to Cognitive Load, items Q13–Q28 correspond to Personalization Alignment, and items Q29–Q40 correspond to Reader Attitude. Dimension scores are obtained by averaging the corresponding item ratings, after which the same dynamic weighting mechanism as in the LLM-based evaluation is applied.

C.5 Ethics Statement

All participants provided informed consent for the use of their anonymized responses for research purposes; guardian consent was obtained for minors. Each participant received compensation of RMB 17.50 for their participation. The human evaluation protocol was reviewed and approved by the Medical Ethics Committee of Tongji Medical College of Huazhong University of Science Technology.

D Metrics and Dimensions

Detailed definitions of PSCB personalization metrics and dimensions is shown in Table 27.

E Significance Tests

E.1 Personalization

We aggregate the three scientific domains and retain only topics evaluated for all three systems. Following paired testing practice for common NLP test instances, we first conduct a Friedman test across CWF, Qwen2.5-14B, and CWF_R. The omnibus differences are significant for child, teen, and adult readers.

All nine pairwise comparisons remain statistically significant after correction ($p_{\text{Holm}} < 0.05$). CWF outperforms Qwen2.5-14B across all reader groups, whereas CWF_R is consistently lower than CWF. Compared with Qwen2.5-14B, CWF_R is higher for teen and adult readers but lower for child readers, indicating an audience-dependent personalization–reliability trade-off.

Table 28 reports the adjusted results without repeating the method means already presented in Table 4.

E.2 Fact-checking

For each domain–reader condition, we compare the article-level factual accuracy scores of CWF_R and CWF on the same evaluation topics. For each matched topic i , we define the paired difference as $d_i = S_{i,\text{CWF}_R} - S_{i,\text{CWF}}$ and conduct a two-sided paired t -test of the null hypothesis $H_0 : E[d_i] = 0$. Thus, a positive t -statistic indicates higher factual accuracy after revision.

Table 29 reports the unadjusted p -values. CWF_R obtains significantly higher factual accuracy in seven of the nine domain–reader conditions at the conventional $p < 0.05$ level. The estimated differences are also positive for AI–Child ($p = 0.0782$) and Medicine–Child ($p = 0.0941$), but neither reaches the conventional significance threshold.

Part	Dimension	Questionnaire Items
A	Basic Information	Q1: Age. Q2: Gender. Q3: Highest education level. Q4: Major field of study (if applicable). Q5: Whether the participant frequently reads science-popularization articles. Q6: Main channels for obtaining science information (multiple choice).
B	B1: INTR	Q1: Understanding the topic of this article requires substantial mental effort from me. Q2: The overall difficulty of this article is appropriate for me. Q3: The amount of information in this article is just right for me, neither too much nor too little. Q4: I gained a considerable amount of new knowledge from reading this article.
	B2: EXTR	Q5: To understand the concepts or terms in the article, I need extra thinking or inference. Q6: The paragraph structure and logical order of the article make it easy for me to follow. Q7: The wording and sentence expression of the article make it smooth to read. Q8: The article is concise and focused, without redundant or off-topic content.
	B3: GERM	Q9: The article helps me understand the key knowledge points through effective devices such as analogies, examples, or summaries. Q10: The explanations and examples provided in the article are sufficient for me to retell its core content. Q11: The article noticeably deepens and expands my understanding of the topic. Q12: Through appropriate summarization and organization, the article helps me form a clear overall mental picture of the topic.
C	C1: CONT	Q13: The scientific aspects emphasized in the article are exactly the parts I care about or am interested in. Q14: The perspective and content selected by the article are practically meaningful to me. Q15: The article contains a large amount of content that feels irrelevant to me. Q16: The content direction covered by the article is highly consistent with what I genuinely want to know.
	C2: KNOW	Q17: The explanation depth and difficulty level of the article fit my current knowledge level. Q18: The article spends too much space explaining things I already know, making the information level lower than I expected. Q19: The information density of the article is just right for me. Q20: I am satisfied with how thoroughly the article explains the key concepts.
	C3: STYLE	Q21: The language style of the article, including vocabulary and tone, feels natural and approachable to me. Q22: The organizational structure of the article makes it easy for me to follow the author's line of thought. Q23: The tone of the article makes me uncomfortable, for example because it is too academic, too childish, or too preachy. Q24: The overall pacing of the article matches my reading habits.
	C4: CONTX	Q25: The examples in the article come from everyday situations familiar to me. Q26: The analogies used in the article feel relatable and help me understand abstract concepts. Q27: The article makes me feel that the scientific knowledge is connected to my own life. Q28: The examples or scenarios in the article feel unfamiliar to me and make it hard to relate to the content.
D	D1: ENG	Q29: The beginning of the article quickly captured my attention. Q30: Throughout the reading process, I remained curious about the content. Q31: The article is vivid and interesting rather than dull. Q32: While reading, I felt like skipping some paragraphs at several points.
	D2: TRU	Q33: I believe the scientific knowledge presented in the article is reliable. Q34: The article maintains appropriate caution when discussing uncertain content and does not exaggerate. Q35: While conveying scientific knowledge, the article does not make me feel forced to accept its claims. Q36: Some expressions in the article make me doubt its scientific accuracy.
	D3: CONTI	Q37: After reading, I became more interested in learning further about this topic. Q38: I would be willing to continue reading similar articles on this topic in the future. Q39: I would be willing to recommend this article to friends with backgrounds similar to mine. Q40: After reading this article, I feel there is no need to further learn about this topic.

Table 26: Paper-formatted version of the Credamo questionnaire used in human evaluation. Negatively worded items are reverse-coded during score aggregation.

Metric	Item	Definition
CL	Overall	Cognitive Load evaluates whether a popular-science article imposes an appropriate level of cognitive demand on the target reader. The goal is not to minimize difficulty indiscriminately, but to calibrate complexity to the reader’s cognitive level: a good article should neither overwhelm the reader with excessive conceptual difficulty nor oversimplify the content to the point of losing educational value.
	INTR	Intrinsic Fit measures whether the conceptual difficulty, information density, and prerequisite assumptions of the article are appropriate for the target persona. High-scoring articles present the right amount of scientific content for the reader’s prior knowledge and cognitive level, without being either overly technical or trivially shallow.
	EXTR	Extraneous Burden Control measures whether the article minimizes unnecessary cognitive burden caused by poor wording, confusing structure, unexplained jargon, redundant details, or abrupt transitions. High-scoring articles are easy to follow and do not impose avoidable processing costs unrelated to the scientific content itself.
	GERM	Germane Support measures whether the article actively facilitates understanding through pedagogically helpful devices, such as analogies, examples, step-by-step explanation, local summaries, and concept scaffolding. High-scoring articles do not merely simplify content; they help the reader construct useful mental schemas for understanding the topic.
PA	Overall	Personalization Alignment evaluates whether the generated article is genuinely tailored to the target reader, rather than merely being generally readable or superficially simplified. In PSCB, personalization includes not only tone adaptation, but also whether the article selects suitable content, explanation strategies, and contextual examples for a specific persona.
	CONT	Content Relevance measures whether the article selects and emphasizes scientific content that is relevant to the target persona’s interests, goals, and likely concerns. High-scoring articles focus on aspects of the topic that matter to the intended reader, rather than presenting generic information indiscriminately.
	KNOW	Knowledge-Level Fit measures whether the explanation depth, abstraction level, and technical granularity match the target persona’s educational background and expected prior knowledge. High-scoring articles explain enough to be informative without assuming too much or talking down to the reader.
	STYLE	Style Consistency measures whether the article’s language style, vocabulary, sentence rhythm, and explanatory tone are consistent with the target persona. For example, writing for children should use simpler wording and more vivid explanations, whereas writing for adult readers may allow denser reasoning and more precise terminology.
	CONTX	Contextual Resonance measures whether the article uses examples, analogies, scenarios, or applications that resonate with the reader’s everyday life or likely experiences. High-scoring articles make the science feel personally meaningful and easier to relate to.
RA	Overall	Reader Attitude evaluates the target reader’s likely affective and behavioral response to the generated article. In personalized science communication, success is determined not only by whether the article is understandable, but also by whether the reader feels interested, trusts the explanation, and is willing to continue engaging with the content.
	ENG	Engagement Appeal measures whether the article is interesting, vivid, and capable of sustaining the reader’s attention. High-scoring articles avoid dry or overly mechanical exposition and instead maintain the reader’s curiosity throughout the text.
	TRU	Trust and Credibility measures whether the article gives the reader a sense of reliability, seriousness, and epistemic confidence. High-scoring articles present knowledge in a way that appears careful, balanced, and scientifically grounded, thereby increasing reader trust.
	CONTI	Continuance Intention measures whether the article encourages the reader to keep reading, explore related content, or further engage with the topic. High-scoring articles leave the reader with the impression that continuing to learn from this source would be worthwhile.

Table 27: Detailed definitions of PSCB personalization metrics and dimensions. CL denotes Cognitive Load, PA denotes Personalization Alignment, and RA denotes Reader Attitude.

Group	Comparison	p_{Holm}	$ d_z $	Sig.
Child	CWF vs Qwen2.5-14B	0.0060	0.230	**
	CWF _R vs Qwen2.5-14B	0.0016	0.283	**
	CWF _R vs CWF	< 0.0001	0.381	***
Teen	CWF vs Qwen2.5-14B	0.0032	0.275	**
	CWF _R vs Qwen2.5-14B	0.0160	0.220	*
	CWF _R vs CWF	0.0160	0.210	*
Adult	CWF vs Qwen2.5-14B	< 0.0001	0.509	***
	CWF _R vs Qwen2.5-14B	0.0090	0.220	**
	CWF _R vs CWF	0.0009	0.299	***

Table 28: Paired significance tests for LLM personalization scores after aggregating across scientific domains. p_{Holm} denotes the two-sided paired-randomization p -value after Holm correction within each reader group; $|d_z|$ is the magnitude of the paired standardized effect. Significance codes are * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and ns $p \geq 0.05$. Method means are omitted because they are reported in Table 4.

Domain	Group	Comparison	t -statistic	p -value	Sig.
AI	Adult	CWF _R vs CWF	2.145	0.0425	*
	Teens	CWF _R vs CWF	3.210	0.0035	**
	Child	CWF _R vs CWF	1.856	0.0782	†
Biology	Adult	CWF _R vs CWF	3.942	0.0008	***
	Teens	CWF _R vs CWF	2.406	0.0242	*
	Child	CWF _R vs CWF	3.093	0.0051	**
Medicine	Adult	CWF _R vs CWF	2.882	0.0084	**
	Teens	CWF _R vs CWF	3.251	0.0032	**
	Child	CWF _R vs CWF	1.765	0.0941	†

Table 29: Two-sided paired t -tests comparing article-level factual accuracy between CWF_R and CWF on matched evaluation topics. Positive t -statistics indicate higher factual accuracy after revision. Reported p -values are unadjusted. Significance codes are * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$; † denotes $p < 0.1$ but not conventional statistical significance.